NUI Galway
OÉ Gaillimh

Autumn Examinations 2012 / 2013

Exam Code(s)	1SD1/1SD3/1MF1/1IT1
Exam(s)	Higher Diploma in Applied Science (Software Design & Development), Higher Diploma in Applied Science (Software Design & Development) Industry Stream, M.Sc. in Software Design & Development, Master of Information Technology
Module Code(s)	CT874/CT536
Module(s)	Programming I/Object Oriented Programming I
Paper No.	I
External Examiner(s)	Professor Michael O'Boyle
Internal Examiner(s)	Professor Gerard Lyons Dr. Michael Madden * Mr. Seamus Hill
Instructions:	Candidates are required to answer: **Section A (20 Marks)** Answer all 10 parts of section A, This section consists of 10 multiple-choice questions all of which should be attempted. **Answers are to be written on the MCQ sheet provided, NOT on the Examination paper.** **Section B (80 Marks)** Answer Any **Two** Questions.
Duration	2 hours
No. of Pages	6
Department(s)	Information Technology
Requirements	MCQ

Section A (20 Marks)
Attempt all 10 questions
Mark your answers on the MCQ sheet provided

1) Which of the following is a scenario that illustrates the benefit of polymorphism?

   a) A function can be given a meaningful name without having to change it to accommodate to rules against having multiple methods of the same name in a single function.
   b) The extension of a class makes the member variables of the parent available to the class, and saves the time of having to redefine these member variables.
   c) The introduction of a new class to an already functioning framework of classes using inheritance only requires a user to implement the new class by extending a parent class and adding the appropriate method.
   d) The extension of a class does not require the programmer to write a method that is defined in the parent. The class instead can call the parent method whenever it wants to.

2) Assume we have three classes, X Y and Z. Y and Z are siblings and are subclasses of the X class. Which classes have access to a private variable in class X?

   a) X, Y and Z
   b) X and Y only
   c) Y and Z only
   d) X only

3) How would we access the 12th element of the array, x?

   a) x(12);
   b) x(12);
   c) x[11];
   d) x[12];

4) Which of the following methods is not contained within the List class?

   a) hasNext
   b) add
   c) List
   d) remove

5) The String class method, toUpperCase _________________.

   a) Makes lowercase characters in the string uppercase and uppercase characters lowercase.
   b) Makes the first character of the string uppercase and the rest lowercase.
   c) Converts a single character to uppercase.
   d) Converts the entire String to uppercase.

6) An assertion is ________________.

   a) A mechanism to improve the likelihood of catching logical errors during program development.
   b) A synonym for a typical Java statement.
   c) A conclusion or stance that was drawn based on reasoning.
   d) A logical, usually Boolean expression.

7) How could a programmer stick to common practice with respect to initializing class variables through constructors?

   a) Use the same name for the parameter corresponding to the class variable as for the class variable
   b) Use a different name for the parameter corresponding to the class variable as for the class variable
   c) Always include multiple constructors.
   d) Use the keyword, this, as much as possible.

8) What is meant by "returning an object" from a method?

   a) It means you are returning a null value to the caller.
   b) It means returning a primitive type to the caller.
   c) It means returning the object's address to the caller.
   d) It means returning the object's value to the caller.

9) What is the difference between a sentinel-controlled loop and a count-controlled loop?

   a) A count-controlled loop is executed for a fixed number of times while a sentinel-controlled loop is executed infinitely.
   b) A count-controlled loop uses the numerical increment operator ++, while the sentinel-controlled loop uses Boolean expressions.
   c) A count-controlled loop is executed for a fixed number of times while a sentinel-controlled loop is executed repeatedly until a certain value is encountered.
   d) A count-controlled loop does not take user input in its body, while a sentinel-controlled loop does.

10) In computer science, what is meant by garbage collection?

   a) Deallocation of unused memory space.
   b) Going through the program and fixing erroneous code.
   c) Deletion of objects from memory.
   d) Defragmentation of the disk drive.

Section B (80 Marks)
Answer Any TWO Questions

Q 2.

a) Write a Java application to simulate the rolling of two dice. The application should use an object of class *Random* once to roll the first die and again to roll the second die. The sum of the two values should then be calculated and displayed. Each die should have an integer value from 1 to 6 so the sum of the values will vary from 2 to 12. Your application should roll the dice 1,000 times. Use a one-dimensional array to count the number of times each possible sum appears. The application should display the sum and frequency in tabular format.

[20]

b) Write a Java application that prints the following diamond shape.

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

[20]

Q 3.

a) In relation to the visibility of data members, discuss the effects of the modifiers *protected*, *public* and *private*, using examples where appropriate.

[10]

b) Write a program that accepts the unit of weight of a bag of coffee in pounds and the number of bags sold and display the total price of the sale, compute as follows:

totalPrice = bagWeight * numberOfBags * pricePerLb;
totalPriceWithTax = totalPrice + totalPrice * taxRate;

Display the results in the following manner:

Number of bags sold: 25
Weight per bag: 5
Price per pound: €5.55
Sales tax: 10.0%
Total price: € 763.13

You are required to define and use ***a programmer-defined*** CoffeeBag class. Include class constants for the price per pound and tax rate with the value of €5.55 per pound and 10% respectively.

[30]

Q 4.

ACME Bank, have requested the development of an application to maintain their customers savings accounts.

You are required to:

- Create a ***SavingsAccount*** class. Use a static variable *annualInterestRate* to store the annual interest rate for all account holders. Each object of the class contains a private instance variable *savingsBalance* indicating the amount the saver currently has on deposit.

[10]

- Provide a method, *calculateMonthlyInterest*, to calculate the monthly interest by multiplying the *savingsBalance* by *annualInterestRate* divided by 12. This interest should be added to *savingsBalance*.

[5]

- Provide a static method *modifyInterestRate* that sets the *annualInterestRate* to a new value.

[5]

- Write a program to test class ***SavingsAccount***. Instantiate two *savingsAccount* objects, *saver1* and *saver2*, with balances of €1,000.00 and €5,000.00 respectively. Set *annualInterestRate* to 4%, then calculate the monthly interest for each of the 12 months and print the new balances for both savers. Then, set the *annualInterestRate* to 5%, calculate the next month's interest and print the new balances for both savers.

[20]

Q 5.

a) Describe how polymorphism promotes extensibility?

[10]

b) Write a Java application that uses *String* method *compareTo* to compare two strings inputted by the user. Output whether the first string is less than, equal to or greater than the second.

[10]

c) Write a Java application that uses a random-number generation to create sentences. Use **four** arrays of strings called *article*, *noun*, *verb*, and *preposition*.

Create a sentence by selecting a word from each array in the following order: article, noun, verb, preposition, article, noun. As each word is picked, concatenate it to the previous word in the sentence. The words should be separated by spaces.

When the final sentence is output, it should start with a capital letter and end with a full stop. The application should generate and display 20 sentences.

- The *article* array should contain the articles “the”, “a”, “one”, “some”, “any”;
- the *noun* array should contain the nouns “cat”, “dog”, “man”, “pub”, “street”;
- the *verb* array should contain the verbs “walked”, “ran”, “crawled”, “strolled”, “ambled”;
- the *proposition* array should contain the prepositions “to”, “from”, “under”, and “on”.

[20]